

SECTION 1. IDENTIFICATION OF THE CHEMICAL AND COMPANY

Product name : RESTORA
 Other means of identification : Not applicable
 Recommended use : Laundry detergent
 Restrictions on use : Reserved for industrial and professional use.
 Product dilution information : Product is sold ready to use.
 Company : Chile ECOLAB S.A.
 Avenida Pedro de Valdivia 3801
 Santiago, Santiago Chile
 (2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600
 sac.chile@ecolab.com
 Emergency telephone : (+56-2) 2635-3600 (CITUC) EMERGENCY PHONE
 Toxicological Emergency Phone : CITUC (+56-2) 2635-3800 (24 hours) Toxicological Emergencies

SECTION 2. HAZARDS IDENTIFICATION

Classification according to NCh 382 : Not classified according to NCh 382

Distintivo según NCh 2190 : Not classified according to NCh 382

GHS Classification

Acute toxicity (Oral) : Category 5
 Skin irritation : Category 2
 Serious eye damage : Category 1
 Short-term (acute) aquatic hazard : Category 3

GHS label elements

Hazard pictograms :



Signal Word : Danger

Hazard Statements : May be harmful if swallowed.
 Causes skin irritation.
 Causes serious eye damage.
 Harmful to aquatic life.

Precautionary Statements : **Prevention:**
 Wash skin thoroughly after handling. Avoid release to the environment. Wear protective gloves/ eye protection/ face protection.

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Response:

IF ON SKIN: Wash with plenty of water. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor. Call a POISON CENTER/ doctor if you feel unwell. If skin irritation occurs: Get medical advice/ attention. Take off contaminated clothing and wash it before reuse.

Disposal:

Dispose of contents/ container to an approved waste disposal plant.

Other hazards : None known.

SECTION 3. COMPOSITION/INFORMATION OF INGREDIENTS

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Oxirane, 2-methyl-, polymer with oxirane, mono(2-ethylhexyl) ether	64366-70-7	30 - 60
2-(2-hexyloxyethoxy)ethanol	112-59-4	10 - 30
alcohols, c12-14-secondary, ethoxylated	84133-50-6	10 - 30
2-phenoxyethanol	122-99-6	5 - 10
Tripropylene glycol monomethyl ether	25498-49-1	5 - 10
[(butoxymethylethoxy)methylethoxy]propan-1-ol	55934-93-5	5 - 10

SECTION 4. FIRST AID MEASURES

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Get medical attention if irritation develops and persists.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Expected acute effects

Eyes : Causes serious eye damage.

Skin : Causes skin irritation.

Ingestion : May be harmful if swallowed.

Inhalation : Health injuries are not known or expected under normal use.

Expected chronic effects

Chronic Exposure : Health injuries are not known or expected under normal use.

Most important symptoms / effects

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Eye contact	: Redness, Pain, Corrosion
Skin contact	: Redness, Irritation, slight irritation
Ingestion	: No information available.
Inhalation	: No symptoms known or expected.

SECTION 5. FIREFIGHTING MEASURES

Extinguishing Agents	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing agents	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Products generated in combustion and thermal degradation	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NO _x) Sulfur oxides Oxides of phosphorus
Precautions for Emergency Personnel and / or Firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. MEASURES IN CASE OF ACCIDENTAL SPILLAGE

Personal precautions, protective equipment and emergency procedures	: Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Handling

Advice on safe handling	: Do not ingest. Avoid contact with skin and eyes. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Wash hands thoroughly after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
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Storage

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Recommended and unsuitable packaging by supplier : Use high density plastic containers.

Storage temperature : 0 °C to 50 °C

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Impervious gloves, resistant to chemicals.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.
Wash face, hands and any exposed skin thoroughly after handling.
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid
Color : clear, colorless
Odor : slight
pH : 7,0 - 9,7, (100 %)
Flash point : Not applicable, Does not sustain combustion.
Odor Threshold : No data available
Melting point/freezing point : No data available
Initial boiling point and boiling range : No data available
Evaporation rate : No data available
Flammability (solid, gas) : Not applicable

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Upper explosion limit	: No data available
Lower explosion limit	: No data available
Vapor pressure	: No data available
Relative vapor density	: No data available
Relative density	: 0,99 - 1,02
Water solubility	: soluble
Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: No data available
Autoignition temperature	: No data available
Thermal decomposition	: No data available
Viscosity, kinematic	: No data available
Explosive properties	: No data available
Oxidizing properties	: The substance or mixture is not classified as oxidizing.
Molecular weight	: No data available
VOC	: No data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: None known.
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides Nitrogen oxides (NO _x) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Toxicity

Product

Acute oral toxicity	: Acute toxicity estimate : 2.995 mg/kg
Acute inhalation toxicity	: No data available
Acute dermal toxicity	: Acute toxicity estimate : > 5.000 mg/kg
Skin corrosion/irritation	: No data available

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Serious eye damage/eye irritation	: No data available
Respiratory or skin sensitization	: No data available
Carcinogenicity	: No data available
Reproductive effects	: No data available
Germ cell mutagenicity	: No data available
Teratogenicity	: No data available
STOT-single exposure	: No data available
STOT-repeated exposure	: No data available
Aspiration toxicity	: No data available

SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity

Environmental Effects	: Harmful to aquatic life.
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Product

Toxicity to fish	: No data available
Toxicity to daphnia and other aquatic invertebrates	: No data available
Toxicity to algae	: No data available

Components

Toxicity to fish	: alcohols, c12-14-secondary, ethoxylated 96 h LC50 Pimephales promelas: 3,5 mg/l Test substance: Information given is based on data obtained from similar substances. 2-phenoxyethanol 96 h LC50 Fish: > 220 mg/l
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Components

Toxicity to daphnia and other aquatic invertebrates	: Oxirane, 2-methyl-, polymer with oxirane, mono(2-ethylhexyl) ether 48 h EC50 Daphnia magna (Water flea): 36,6 mg/l 2-(2-hexyloxyethoxy)ethanol 48 h EC50 Daphnia: 433 mg/l alcohols, c12-14-secondary, ethoxylated 48 h EC50 Daphnia magna (Water flea): 3,1 mg/l Test substance: Information given is based on data obtained from similar substances. [(butoxymethylethoxy)methylethoxy]propan-1-ol 48 h LC50: > 100 mg/l
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Components

Toxicity to algae	: Oxirane, 2-methyl-, polymer with oxirane, mono(2-ethylhexyl) ether 72 h EC50 Desmodesmus subspicatus (green algae): 31,9 mg/l Tripropylene glycol monomethyl ether
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96 h EC50: 9.069 mg/l

Persistence and degradability

Biodegradable

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. INFORMATION OF DISPOSAL

Disposal of contaminated packaging : In accordance with what is described by Supreme Decree No. 148, the packaging of the product is considered hazardous waste and must be disposed of through companies authorized to receive and / or treat such waste, which must be issued and final disposal certificate waste.

SECTION 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

Official transport classification of the United Nations

Not dangerous goods

Air transport (IATA)

Official transport classification of the United Nations

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

Official transport classification of the United Nations

Not dangerous goods

SECTION 15. REGULATORY INFORMATION

Registrations and Certifications

Chile: Our HDS meets Chilean Normative: NCH 2245: 2015 (Safety Data Sheet for chemical products - Content and order of sections)

The receiver should pay attention to the possible existence of local regulations.

NCh 1411: Prevention of risks, IV identification of risks of materials

NCh 2190: Transportation of Hazardous Substances - Pictograms for hazard identification

NCh 382: Dangerous Good - Classification

D.S. No. 594: Minimum basic conditions in the workplace

D.S. No. 148: Hazardous waste disposal

D.S. No. 72: Mining Safety Regulations

D. S. No. 43: reports on the storage of hazardous substances

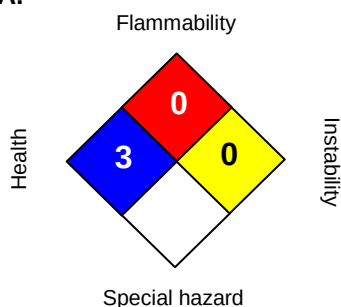
D. S. No. 40: Reporting on exposure hazards

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SECTION 16. OTHER INFORMATION

NFPA:



HMIS III:

HEALTH	3
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 07.07.2020
Version : 1.0
Prepared by : Regulatory Affairs

While there is no change in the formula or hazard classifications, this SDS remains valid.

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.